

aspect ratio CNT reinforcement. PE-0.05HARC shows 13% and 29 % improvement in hardness and elastic modulus as compared to pure PE. But, in case of PE-0.05LARC, a smaller improvement of 3% in hardness and 8% in elastic modulus is recorded. These results clearly indicate that HARCs have an edge over LARCs in terms of mechanical behavior of the composite. Same trend is observed in case of composite with 0.1 wt. % nanofillers. PE-0.1HARCs show an impressive improvement of 45% in hardness and 42% in modulus, whereas the improvement in case of PE-0.1LARC is 19% in hardness and 28% in elastic modulus over pure PE. Appreciable improvement in mechanical properties for composites reinforced with HARCs can be explained in terms of their high surface area to volume ratio and interfacial strength. These factors increase the load transfer efficiency to the CNT network throughout the composite uniformly and avoid the failure due to high stress concentration. Cipiriano et al. [48] have also noticed positive effect of aspect ratio of CNT on storage modulus, elastic modulus and viscoelastic property of polystyrene/CNT composite. HARCs, in the present study, have diameter 10-12 nm, which is in much closer range to typical polymer chains with diameter range of $\sim 1-3$ nm [49]. This opens up two possibilities. Firstly, the HARCs can align themselves along polymer chains, increasing a better interfacial bond area and thus more effective load transfer. Secondly, due to the similar diameter and the broken C-C bonds present in CNT, they can act as cross linkers between polymer chains and making the inter-chain sliding difficult. This in turn restricts the plastic deformation and increases the strength of the polymer. In case of LARCs, due to their huge mismatch in size (diameter) with the polymer, they cannot align or act as cross-linkers. Rather, they are potential cause of disturbance in the arrangement of polymer chains, decreasing the strength. This hypothesis is explained with the help of a schematic presented in Fig.8.

Further, at identical composition level, the high aspect ratio CNTs have more number of total floccs as compared to low aspect ratio CNTs, resulting in percolation network with compact structure. This increases the micro mechanical interlocking of CNT and polymer chains, holding the matrix together and restricting the movement of polymer chains through bridging between chains. In case of LARCs, chances of interfacial bonding and mechanical locking are lesser due to their larger diameter and shorter length. Thus, the mechanisms of strengthening in HARCs are manifold, as compared to LARCs with only sharing of load through interface. This explains the significantly higher hardness and elastic modulus in polymer matrix with HARC reinforcement as compared to LARCs. The reinforcement potential of fibers, as a function of their aspect ratio and volume fraction, has been explained by some theoretical models. Amongst these, shear lag based models and Halpin-Tsai model are widely used. A modified

form of Halpin Tsai relationship, as proposed by Kanagarj et al. [43] and Coleman et al. [50], gives a simpler approach getting rid of information related to the thickness of interfacial region and fiber alignment direction. Thus, this one is more suitable for present study with randomly oriented nanotubes well adhered in the polymer matrix. In this model, both volume fraction and aspect ratio of CNT are considered to estimate the elastic modulus, as following:

$$E_C = (\eta_1 \eta_o E_{CNT} - E_{PE}) V_{CNT} + E_{PE} \quad (4)$$

$$\text{where, } \eta_1 = \text{length efficiency factor} = 1 - \frac{\tanh(a - 1/d)}{(a - 1/d)}, \quad a = \sqrt{\frac{-3E_{PE}}{2E_{CNT} \ln(V_{CNT})}}$$

η_o = Orientation efficiency factor = 0.2 for randomly oriented fiber; Young's modulus of CNT is 910 GPa [51] and for PE is 0.95 GPa (from the present study).

The above expression is used for estimating the theoretical elastic modulus of different compositions used in this study. Fig .9 shows comparison of experimental and theoretical elastic modulus for the two selected aspect ratios of CNTs. It can be observed that modified Halpin-Tsai model predicted values are in the similar range to that of the experimental values. Further, regression analysis is done to verify how this experimental result fits to the theoretical results. R^2 (coefficient of determination) values are calculated for both HARC and LARC in different volume fraction. The R^2 value for HARC and LARC is 0.822 and 0.957, respectively, which indicates linear relationship between the predicted theoretical and experimental results of the composite. Halpin Tsai equation fits very well for low volume fraction. But linear increase of modulus is predicted for higher volume fraction [50] as shown in fig.9. Thus, modified Halpin-Tsai model is a better fit for this study, as the volume fraction of reinforcement is < 0.1 wt. % in all cases. High aspect ratio CNTs have shown high modulus because a good fraction of isolated nanotubes are potentially aligned with polymer chains, which leads to high interfacial bond and cross linking of polymer chains. Fig. 8 clearly shows HARC aligned along the direction of polymer chains in fracture surface of the composite. The strong coating of polymer around the CNT and their interfacial bonding can easily transfer the load on matrix to CNTs and help in avoiding the rupturing of polymer chains as well as in minimizing the stress concentration in the composite.

3.4 Effect of Aspect Ratio of CNT on Tribological Properties

Fig.10 shows the change in co-efficient of friction (COF) as a function of sliding distance at different compositions of UHMWPE-CNT composite. The COF increases from zero to peak value during running-in-period due to the influence of surface roughness and thereafter attains a steady state. It is observed that addition of CNTs to polymer remarkably reduces the friction force as compared to pure PE composite. One of the key reasons is the role played by carbon nanotubes as solid lubricants, by releasing graphene layers on wear track [52]. MWCNTs possess concentric cylindrical layers of graphene, bonded to each other with Vander waals forces [53]. Individual cylinders of multi wall carbon nanotubes are expected to slide or rotate easily with respect to each other because MWCNT interlayer corrugation energy is small. Therefore, it is expected that decreased friction is due to worn off CNTs acting as a lubricants, like solid powder graphite lubricants [35, 52-54]. In addition, the open part of the reinforced CNTs, which are protruding and dangling on surface, can also act as lubricator due to their smooth cylindrical surface that gives a rolling action. Aspect ratio of the CNTs greatly influenced the friction force irrespective of the concentration. HARC has shown 120 and 25 % lesser COF as compared to LARC at 0.1 and 0.05 wt. % of concentration, respectively (Fig.10). The percolation network formation of CNTs strongly depends on the aspect ratio. It is expected that the HARC is rooted deeply and strongly in the polymer matrix and it contributes to continuous lubrication effects. Fracture surfaces of the composites have also shown longer HARC protruding out of matrix as compared to LARC (Fig. 3). This in turn signifies the presence of CNTs in wear track for longer time to be able to release more graphene layers. On the contrary, LARC tends to break from the polymer matrix and get detached from the track. Thus, they cannot provide as efficient lubrication as HARC.

The surface profiles of the wear tracks were measured in the transverse direction of the sliding. Fig.11 furnishes the representative profiles of wear tracks in each composition. The average width and depth of the PE track (2.5mm and 35 μ m respectively) are much higher than that of PE-0.05LARC (2mm and 30 μ m) and PE-0.05HARC (1.5mm and 17 μ m). The width and depth of the track on PE-0.1LARC is 1.5mm and 20 μ m, respectively as compared to those in PE-0.1HARC being 0.7mm and 4 μ m. Furthermore, the width and depth of the track decreased remarkably with increasing CNT concentration. All together, this shows significant improvement in wear resistance with CNT concentration, as well as increasing aspect ratio of CNTs.

The variation in specific wear rate with composition is shown in Fig 12. Addition of 0.05 wt. % of HARC and LARC reduces the wear rate by 76% and 18%, as compared to PE. Addition of 0.1 wt. % reinforcement reduces the

wear rate by 530% and 84%, respectively, for HARC and LARC. This enormous improvement in wear resistance is expected to be a combined effect of increase in mechanical properties and decrease in COF.

The worn surfaces of the prepared composites were examined to understand the predominant wear mechanism (Fig. 13). Continuous sliding causes increased polymer asperities at the tribocontact with the harder counter body. Further, due to rubbing action, the elastic plastic deformation of polymer asperity increases the deep furrows and it further rolls into thin molecular sheets. This results in deeper grooves and generates wear debris on the polymer surface. Deep craters and damages on the rough wear track of PE (Fig. 13a) indicate low wear resistance of this surface. Hard asperities and compaction of wear debris (Fig.13a') lead to increase in COF. But, in case of CNTs reinforced composite, smoother wear track surface without broad exfoliation and microcracking (Fig.13 b and c) is observed. Strong interfacial shear resistance of the CNT reinforced composite restricts the ploughing and cutting mechanism. The effective load transfer from polymer to CNTs suppresses the plastic deformation and cracking, which leads to less removal of the polymer particles from the composite. In addition, continuous lubrication effect from CNTs results into a smoother worn surface. CNTs get exposed on the surface of the composite during sliding, creating the interface and preventing direct contact between the counter body surfaces, which greatly reduce the ploughing and scratching effect in the composite. This effect is predominant in case of HARC, because these CNTs are deeply rooted within the matrix (Fig 13c'). On the other hand, LARCs come out with the polymer coating. Thus, they fail to give continuous lubrication effect and also increases the debris in the surface (Fig.13b').

While comparing the effect of CNT aspect ratio on wear behavior, HARC is found to show significant improvement over LARC at both the concentration. HARC composite (Fig 11) shows shallower and smoother wear track with less width and depth as compared to LARC composite. Smoother wear tracks indicate that the tribological loss is mainly governed by adhesive wear, which results in less wear volume. The effect is more prevalent with increasing nanofiller content, noting an impressive decrease in specific wear rate of 49% and 243% at 0.05 and 0.1 wt. % of HARC content, respectively, as compared to similar concentrations of LARCs (Fig. 12). The additional improvement in wear resistance for HARC reinforced composite (w.r.t. LARCs) can be explained in terms of better mechanical behavior and lower COF in the former. The strong interfacial bonding of HARC to polymer chains and their self entangled compact structure prevents the easy removal of debris and relative movement of polymer chains. All the above discussions point towards the benefit of using HARC as reinforcement in PE composite for improved tribological behavior.

4. CONCLUSIONS

In the present study, CNTs with two different aspect ratios (HAR-900 and LAR-75) were reinforced with PE matrix. Composite having higher aspect ratio of CNTs has shown greater hardness and elastic modulus as compared to CNTs with lower aspect ratio. Differential improvement in mechanical properties of composite with HARC and LARC is explained in terms of high surface area to volume ratio and strong interfacial bonding in the former, which increases the load transfer ability from matrix to CNT. Thermal stability of the composite containing HARC is also better as compared to LARC. Tribological behavior of the composites reveals significantly lower coefficient of friction and specific wear rate for HARC reinforced composites, which is the ideal requirement for its intended application in orthopedic joints. Better attachment of HARC to polymer matrix and their morphological similarity to polymer chains are found to be important factors in fortifying their dual role as toughener and solid lubricator for polymer matrix.

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